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Dynamic reality modeling for AECO applications

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DYNAMIC REALITY MODELING FOR AECO APPLICATIONS

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Image capture process

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Chapter 4

Alignment And Updating

4.1 Introduction

The AECO industry needs up-to-date and complete geo-spacial data of their facilities, ranging from construction site monitoring to immersive visualisations. Despite an enormous amount of data being captured, this goal remains a great challenge due to several factors. First, the captured data is often incomplete due to occlusions or limited sensor range. Second, captured data quickly becomes outdated, while making numerous consecutive full recordings is very time-consuming and expensive. Finally, the data coming from lower-end, cheaper capture devices is rarely used due to a lack of a proper framework to incorporate them into larger datasets. A way to combat these shortcomings is to make better use of the lower-end data. While these sensors lack range and precision, the increased recording speed and lighter data load is a big advantage over traditional devices. As shown in figure **??**, combining high-end with lower-end datasets, by updating small parts of the full point cloud, could extend their relevance and decrease the total recording investment.

Using XR devices like the Hololens 2 as the lower-end sensor can bring additional advantages because they can capture both 2D localised Images and 3D meshes, and get realtime coverage feedback. Allowing users to seek out occlusions and inconsistencies of the previously captured high-end dataset on-site.

Existing works rely on using multiple sensors to capture data at different scales and levels of detail, ranging from adding highly detailed scans of small selected objects [**?**], to filling in large occluded areas using drone footage[**?**]. These methods, however, rely on a (semi-)manual alignment step, making the quick

feedback loop from an XR device not feasible. Merging different datasets in the current state of the art is mostly focused on improving the coverage of the initial scan [**?**]. Updating the initial point cloud with supplementary data captured at a different point in time is a field of little research.

The goal of this work is to set up a pipeline to complement and update existing point clouds using automatically aligned XR datasets captured on site. This will keep the dataset relevant over a longer period of time, requiring fewer full-scale recordings and better overall coverage of the site.

4.2 Related work

4.3 Methodology

Before 2 datasets can be combined, as explained in **??**, they need to be aligned. Each recording set is stored as a session, containing all the metadata and raw files. To get feedback on the current state of the scan, the alignment process needs to happen while the user is on-site. This is why an automatic method is proposed in 4.3.1. This relies on a connection between a python webserver and the XR device to access the reference datasets and increased computational performance.

4.3.1 Dataset alignment

XR devices can record both images and meshes, so multiple pose estimations can be made using 2D and 3D matching techniques. These estimations all have matching parameters like the reprojection error, overlap, inliers and sensor type. These normalized values are used as weights to evaluate the confidence of the proposed pose. The final pose is determined as a weighted average of all the estimations.

Image alignment

The 2D images are matched with the reference dataset, using the OpenCV framework to generate ORB features and state-of-the-art 2D matching algorithms to estimate a pose for each captured image [**?**].

Point cloud alignment

The 3D mesh is matched against the point cloud using Open3d FPFH Feature matching, using the Fast global registration algorithm to determine the transformation [**?**]. The point cloud is subdivided into parts of similar size to the mesh to make multiple estimations.

Pose Estimation

The final pose estimation is the weighted average of the different methods and their confidence factor. The accuracy of this method is precise up to the resolution of the sensor as seen in table 4.1 where it was tested on multiple datasets.

4.3.2 Dataset updating

The main challenge is determining which points in the point cloud are still up to date. This includes the removal and addition of the newer XR data onto the existing reference dataset. We propose a 3 step process as outlined in figure 4.1.

Relevant point selection

After the XR dataset is cleaned up, the reference points are sub-selected using the convex hull of the XR dataset to ensure only points within the newly scanned area can be altered. This method is more precise than a bounding box and leads to fewer false positives.

Out-of-date point removal

To determine which points are out of date, each one must pass at least one of the following checks: a coverage or occlusion check. The coverage is determined with a threshold distance query against the cleaned-up XR mesh. The visibility check serves to maintain points that were not visible to the XR device due to occlusions and is determined by comparing the normals of the n closest faces from the flagged point. If the RANSAC filtered majority of faces are facing away from the point, it is flagged as occluded and fails the check.

Table 4.1: The results of the different matching methods, showing the accuracy is up to the resolution of the Hololens sensor

images/Pose_estimation_table.png

New point addition

The reference point cloud is assumed to have much a higher accuracy than the sampled XR data, so only the not covered XR points are added. To determine these points, an inverse distance query is performed compared to 4.3.2. By default, the XR mesh does not contain any colour information, this is retrieved from the localised XR images by determining the corresponding pixel value through a raycast to the new points.

images/mergingWorkflow.png

Figure 4.1: A three-step process for merging datasets with a subselection of relevant reference data(a), the removal of out-of-date points (b) and the addition of new points(c).

4.4 Experiments

4.5 Conclusion

This work has shown a method to update large existing point cloud datasets without the need to recapture the whole site. XR devices provide multi-sensory data that can not only be used to locate the user in a reference point cloud

to get visual feedback of the site but also to update and complete the existing dataset with new, more focused data.

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Conclusion and future works

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Statement on the use of Generative AI

I did use the generative AI assistance tools ... Generative AI was used for generating programming code.
The text, code, and images in this thesis are my own (unless otherwise specified). Generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.

Curriculum Vitae

Jelle Vermandere was born on June 26, 1995. In 2020, he obtained his master's degree in Architectural Engineering at UGent, graduating with distinction. His thesis focussed on creating a website and database for interactive viewing of 3D models withing exhibitions.

After his studies, Jelle was a self-employed freelancer, creating interactive architectural renderings and video games for various clients.

In July 2021, he started his research at the Faculty of engineering technology at the KU Leuven under the supervision of Prof. Vergauwen and Prof. Bassier. A year later, he secured an FWO strategic basic research grand, allowing him to focus his main research which centers around innovating the built environment through Reality Modelling. The goal of this research is to create dynamic and complete 3D models of indoor environments captured using 3D scan-devices.

Jelle has a wide research interest, focussing around the combination of real and digital techniques. Outside his research, his hobbies led him to mastering the full real-to-digital-to-real workflow using laser scanning, 3D modeling and 3D printing, he can create almost anything. He documents his projects on his YouTube channel where he created a community of over 90.000 subscribes accumulating over 7 million views.

Jelle also likes to practice a wide range of sports including: running, biking, swimming and climbing. After all the hours behind his computer, the physical exercise is a very welcome distraction.

Publications

Journal Articles

1. Vermandere, J., Bassier, M., & Vergauwen, M. (2022). *From Partial Scans to Editable Scenes: Hybrid Object and Scene Completion for Dynamic 3D Environments.* **Automation in Construction**, 14(11), 2680. DOI: 10.3390/rs14112680

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